DANILO ANTONIO TARANTINO

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PROFESSIONAL PROFILE

Electronic Engineer specialized in FPGA architectures, digital acceleration, and power electronics, combining deep technical design skills with a proven track record in client-facing roles, B2B partner relations, and on-site problem-solving. Skilled in international coordination and stakeholder communication. Seeking a Field Application Engineer (FAE) or Technical Sales role to bridge complex semiconductor hardware with customer success and business growth.

EDUCATION

Master's Degree in Electronic Engineering — Final Grade: 105/110
University of Calabria (Unical) – Rende (CS), Italy — **October 2022 – July 2026**

Bachelor's Degree in Electronic Engineering
University of Calabria (Unical) – Rende (CS), Italy — **September 2018 – September 2022**

SELECTED HARDWARE & SYSTEM PROJECTS

- **FPGA IP Cores for AI Activation Functions (Master's Thesis Project)**
  - **RTL Architecture & Optimization:** Designed scalable, highly parallel, and pipelined VHDL architectures on Xilinx Artix-7 for Vision Transformers (ViTs), bypassing hardware DSPs via LUT approximation to preserve resources. Explored design-space trade-offs balancing silicon area, routing congestion, and AXI4 bandwidth limits.
  - **Verification & Simulation:** Developed Python golden models (FP64 reference) and fixed-point quantization tables to generate test vectors, validating hardware simulation outputs against MATLAB mathematical models through comparative error analysis.
  - **Debugging & Performance Analysis:** Executed functional and post-layout simulations in Vivado Simulator, analyzing dynamic and static power trade-offs via SAIF/VCD switching activity files.
- **IoT Wireless Intruder Detection & Smart Alarm Infrastructure**
  Built a wireless sensor network using TelosB nodes and TinyOS, focusing on reliable communication protocols, distributed embedded nodes, and real-time event handling.
- **Modular Framework for Hybrid Powertrain Energy Optimization**
  Developed a MATLAB/Simulink simulation framework for hybrid powertrain energy optimization, evaluating conversion efficiency and thermal/electrical performance.

TECHNICAL SKILLS

- **HDLs & Design:** VHDL, Verilog, RTL Design, Pipelining, FSMs, Area & Resource Optimization, AXI4.
- **Programming Languages:** Python, C Language.
- **Tools & Software:** Xilinx Vivado, Cadence Virtuoso, PC1D, LabVIEW, MATLAB/Simulink.
- **Domains & Systems:** Embedded Systems, Microelectronics, ASIC Concepts, Power Electronics, XFC (Extreme Fast Charging) Systems, Energy Conversion, Electronic Bioengineering, Deep Neural Networks, Automotive ADAS, IoT Architectures, Photovoltaics.

PROFESSIONAL & BUSINESS EXPERIENCE

- **Life Events Ltd** — **September 2022 – September 2026**
  - **Team Leadership & Recruiting (2026 - Saranda, Albania - Hybrid - Contract):**
    Coordinated cross-border teams, conducted remote onboarding and training webinars for new hires.
  - **Operations Oversight & B2B Partnerships (2024–2025 - St. Julian's, Malta - Hybrid - Contract):**
    Directed field operations, negotiated B2B agreements with commercial partners, and led rapid on-site troubleshooting to resolve critical operational bottlenecks.
  - **Concierge Specialist (2022–2023 - St. Julian's, Malta - On Site - Seasonal):**
    Proactively engaged prospects in fast-paced environments, driving direct sales while building customized packages (events, boats, vehicles, accommodations) and client support.
- **Hardware Technician & IT Support Specialist (Italy)** — **September 2015 – August 2018**
  Provided direct technical support, PC assembly, hardware diagnostics, and electronic troubleshooting for private and business clients.

LANGUAGES & MOBILITY

- **Languages:** Italian (Native), English (Fluent), Spanish (Basic).
- **Mobility:** Full availability for international relocation and business travel.